

Literature status: separation growth of ℓ_p , $p > 2$

Run `fa_banach_001`; source arXiv:2112.11523

Source question

Naor's *Extension, separation and isomorphic reverse isoperimetry* defines $\text{SEP}^n(M)$ to be the supremum of the separation moduli of subsets of a metric space M with cardinality at most n . In Section 1.7.6, after discussing dimension reduction, it asks the following question for ℓ_p , $p > 2$.

So, if $p \leq 1 + c(\log \log \log n) / \log \log n$ for some universal constant $c > 0$, then Theorem 81 does not improve asymptotically over $\text{SEP}^n(\ell_p) \lesssim \log n$, while Conjecture 82 would imply that $\text{SEP}^n(\ell_p) = o(\log n)$ if and only if $\lim_{n \rightarrow \infty} (p - 1) \log \log n = \infty$.

For fixed $p \in (2, \infty)$, at present we do not see how to obtain an upper bound on $\text{SEP}^n(\ell_p)$ of the form $o_p(\log n)$ as $n \rightarrow \infty$. We state this separately as an interesting and challenging open question.

Question 83. Is it true that for every $n \in \mathbb{N}$ and $p \in (2, \infty)$ we have $\lim_{n \rightarrow \infty} \text{SEP}^n(\ell_p) / \log n = 0$? More ambitiously, is it true that $\text{SEP}^n(\ell_p) \lesssim_p \sqrt{\log n}$?

Note that $\text{SEP}^n(\mathbf{X}) \gtrsim \sqrt{\log n}$ for any infinite-dimensional normed space $\mathbf{X}$, because by Dvoretzky's

Figure 1: arXiv:2112.11523, page 44: Question 83 on separation growth of ℓ_p , $p > 2$.

The ambitious part of the question asks whether, for every fixed $p \in (2, \infty)$,

$$\text{SEP}^n(\ell_p) \lesssim_p \sqrt{\log n}.$$

This would imply the preceding nontrivial-growth question $\text{SEP}^n(\ell_p) / \log n \rightarrow 0$.

Supporting literature status

Naor and Ren's later paper *Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$* explicitly cites this question as [2, Question 83] and proves the sharp fixed- p answer for L_p .

Proof intuition

Question 83 was difficult because for $p \neq 2$ one cannot simply reduce a finite subset of ℓ_p to logarithmic dimension by a Hilbert-style dimension-reduction lemma. Naor–Ren instead prove a direct partition theorem for finite subsets of the ambient L_p space. Since ℓ_p is an isometric subspace of L_p , any finite subset of ℓ_p inherits the same randomized partition bound from the larger space.

The literature answer

Theorem 1. For every fixed $p \in (2, \infty)$,

$$\text{SEP}^n(\ell_p) \lesssim_p \sqrt{\log n}.$$

while if $\liminf_{n \rightarrow \infty} (p-1) \log \log n < \infty$, then $\text{SEP}^n(L_p) \gtrsim \log n$. See [80, Section 1.7.6] for more on this (including sharp bounds for fixed $1 < p \leq 2$), where it is conjectured that $\text{SEP}^n(L_p) = o(\log n)$ if and only if $\lim_{n \rightarrow \infty} (p-1) \log \log n = \infty$.

It was asked in [77, Question 1] (reiterated in [80, Question 83]) if L_p has nontrivial separation growth when $2 < p < \infty$ (see [80, Section 1.7.6] for the relation to metric dimension reduction). This question was answered affirmatively in [48] by showing that $\text{SEP}^n(L_p) \lesssim (\log n)^{1-1/p}$ for every $(p, n) \in (2, \infty) \times \mathbb{N}$. It was also asked in [77, 80] if, in fact, for every $(p, n) \in (2, \infty) \times \mathbb{N}$ we have $\text{SEP}^n(L_p) \lesssim_p \sqrt{\log n}$, which would be asymptotically optimal as $n \rightarrow \infty$ for every fixed $2 < p < \infty$ (see below). Here we prove that this sharp evaluation of the largest possible separation modulus of an n -point subset of L_p indeed holds:

Theorem 1.10. *For every $2 < p < \infty$ and every $n \in \{2, 3, \dots\}$ we have $\text{SEP}^n(L_p) \asymp_p \sqrt{\log n}$. More precisely,*

$$(1.17) \quad \forall (p, n) \in (2, \infty) \times \mathbb{N}, \quad \sqrt{\log n} \lesssim \text{SEP}^n(L_p) \lesssim p^2 \sqrt{\log n}.$$

Figure 2: arXiv:2502.10543, page 11: explicit reference to [2, Question 83] and Theorem 1.10.

- [77] A. Naor. Probabilistic clustering of high dimensional norms. In *Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 690–709. SIAM, 2017.
- [78] A. Naor. Metric dimension reduction: a snapshot of the Ribe program. In *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018. Vol. I. Plenary lectures*, pages 759–837. World Sci. Publ., Hackensack, NJ, 2018.
- [79] A. Naor. Impossibility of almost extension. *Adv. Math.*, 384:Paper No. 107761, 34, 2021.
- [80] A. Naor. *Extension, separation and isomorphic reverse isoperimetry*, volume 11 of *Memoirs of the European Mathematical Society*. European Mathematical Society (EMS), Berlin, 2024.
- [81] A. Naor, V. Rencs, O. Schramm, and S. Sheffield. Markov chains in smooth Banach spaces and Gromov–

Figure 3: arXiv:2502.10543, page 59: bibliography entries [77] and [80], identifying the 2017 SODA question and Naor’s 2024 monograph.

Consequently,

$$\lim_{n \rightarrow \infty} \frac{\text{SEP}^n(\ell_p)}{\log n} = 0.$$

Thus Question 83 of [2] has an affirmative answer for each fixed $p > 2$.

Proof. Naor–Ren’s Theorem 1.10 [3] states that, for every $(p, n) \in (2, \infty) \times \mathbb{N}$,

$$\sqrt{\log n} \lesssim \text{SEP}^n(L_p) \lesssim p^2 \sqrt{\log n}.$$

The upper bound is the needed part.

Embed ℓ_p isometrically into an L_p space, for example by representing a sequence as a step function on pairwise disjoint measurable sets with the usual normalization. If $S \subset \ell_p$ and $|S| \leq n$, then S is also an n -point subset of L_p , and therefore

$$\text{SEP}(S) \leq \text{SEP}^n(L_p) \lesssim p^2 \sqrt{\log n}.$$

Taking the supremum over all such $S \subset \ell_p$ gives

$$\text{SEP}^n(\ell_p) \leq \text{SEP}^n(L_p) \lesssim p^2 \sqrt{\log n}.$$

For fixed p , the right hand side is $o(\log n)$, which also answers the nontrivial-growth part of Question 83. $\square$

Verification notes and limitations

This packet is a literature-status identification, not a new proof from the run. The supporting paper explicitly says that the sharp $\sqrt{\log n}$ question was asked in [1] and [2], and more specifically identifies the latter as [2, Question 83]. Its bibliography identifies [2] as *Extension, separation and isomorphic reverse isoperimetry*.

The answer is for fixed $p > 2$. Naor–Ren note that, when p is allowed to vary with n , the range $\sqrt[4]{\log n} \lesssim p = o(\log n)$ remains open. This packet also does not address the other reverse-isoperimetry, shape-optimization, or Schatten-class conjectures in [2].

Before promotion, the run lightweight indexes were searched for arXiv:2112.11523, the title phrase, “separation growth”, “SEPⁿ”, and related terms. No existing exact packet or attempt was found. The only nearby registry hit was the distinct arXiv:1506.04398 / arXiv:2502.10543 packet on Lipschitz extension upper bounds.

Human review recommendation

Treat arXiv:2112.11523 Question 83 as already answered affirmatively, in its fixed- $p > 2$ and $\sqrt{\log n}$ form, by arXiv:2502.10543. Keep the other questions in the source paper separate.

References

- [1] A. Naor, *Probabilistic clustering of high dimensional norms*, Proceedings of the Twenty-Eighth Annual ACM-SIAM Symposium on Discrete Algorithms, 690–709, SIAM, 2017.
- [2] A. Naor, *Extension, separation and isomorphic reverse isoperimetry*, Memoirs of the European Mathematical Society, vol. 11, European Mathematical Society, Berlin, 2024; arXiv:2112.11523.
- [3] A. Naor and K. Ren, *Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$* , arXiv:2502.10543.